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- BST provided $O(\log n)$ insertion and search for memory segment addresses.
- In-order traversal returned sorted memory layout: 1000, 2000, 3000, 4000, 5000, 7000, 8000, 9000.
- Search for 3000: true | Search for 6000: false (not allocated).
Time Complexity:
- Insert / Search / Delete: $O(\log n)$ average, $O(n)$ worst case.
- Compared to a simple array, BST performs significantly better for dynamic OS memory operations due to its hierarchical sorted structure.

GitHub Link:

<https://github.com/egotpat-gif/DSAIL-Case-Studies/blob/main/CO1-Virtual%20Memory%20Management>